

Physics Department
Fisika Departement

Semester Test 2

Semestertoets 2

FSK116
26 April 2012

Surname and initials MEMO Student number _____
Van en voorletters *Studentenommer*

Signature _____
Handtekening

Time 2½ hours Max. Total 80 Marks Max
Tyd 2½ ure Maks. Totaal 80 Punte Maks

Internal Examiner: Dr J M Nel External Examiner Mr R Q Odendaal
Interne Eksaminator *Eksterne Eksaminator*

• Answer all questions. • Clearly show all calculations. Do not just write down the answer. • Clearly state all assumptions, rules and laws used. • Where applicable draw the free-body diagrams. • Demonstrate clearly that you understand the physics that you use to answer the question. This means that a numerically correct answer will not necessarily be awarded full marks. • Programmable calculators may NOT be used. • No textbooks allowed. • A formula sheet is attached. • All test and examination regulations of the University of Pretoria are applicable.	• <i>Beantwoord alle vrae.</i> • <i>Wys al jou berekeninge. Moenie net 'n antwoord neerskryf nie.</i> • <i>Stel duidelik alle aannames, reëls en wette wat u gebruik.</i> • <i>Waar van toepassing, teken die vryliggaamsdiagram (of kragte diagramme).</i> • <i>Wys dat jy die fisika wat jy gebruik om die vraag te beantwoord verstaan. Dit beteken dat volpunte sal nie noodwendig vir 'n numeries-korrekte antwoord gegee word nie.</i> • <i>Programmeerbare sakrekenaars MAG NIE gebruik word nie.</i> • <i>Geen handboeke word toegelaat nie.</i> • <i>'n Formuleblad is aangeheg.</i> • <i>Alle toets- en eksamenregulasies van die Universiteit van Pretoria is van toepassing.</i>
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Surname and initials _____ Student number _____
Van en voorletters *Studentenommer*

Signature _____ Venue _____
Handtekening *Lokaal*

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	$v^2 = v_o^2 + 2a(x - x_o)$	$R = \frac{v_o^2}{g} \sin 2\theta_o$
$v_{avg} = \frac{1}{2}(v_o + v)$	$x - x_o = \frac{1}{2}(v_o + v)t$	$T = \frac{2v_o \sin \theta_o}{g}$
$v = v_o + at$	$a = \frac{v^2}{r}$	$f = \mu N$
$x = x_o + v_o t + \frac{1}{2}at^2$	$g = 9.8 \text{ m/s}^2$	$D = \frac{1}{2}C \rho A v^2$
$x = x_o + v_{avg} t$	$h = \frac{v_o^2 \sin^2 \theta_o}{2g}$	
$x - x_o = vt - \frac{1}{2}at^2$		

ALL STEPS MUST TO BE SHOWN. DO NOT JUST WRITE DOWN THE ANSWER. DRAW ALL FREE BODY DIAGRAMS.

ALLE STAPPE MOET AANGEDUI WORD. MOET NIE NET DIE ANTWOORD NEER SKRYF NIE. TEKEN ALLE VRY-LIGAAMSDIAGRAMME.

1 Question 1 / Vraag 1 (ch 2Q20)

1.1 The position of a particle is given by $x = 20t - 5t^3$, where x is in meters and t is in seconds.

Die posisie van 'n deeltjie word gegee deur $x = 20t - 5t^3$, waar x in meter is en t in sekondes.

a. When, if ever, is the particle stationary?

Wanneer, indien ooit, staan die deeltjie stil?

[3]

$$v = \frac{dx}{dt} = 20 - 15t^2$$

When $v = 0 \text{ m/s}$, $t = \sqrt{\frac{20}{15}} = 1.16 \text{ s}$

b. When is the particle's acceleration zero?

Wanneer is die deeltjie se versnelling nul?

[3]

$$a = \frac{dv}{dt} = -30t$$

When $a = 0 \text{ m/s}^2$, $t = 0 \text{ s}$

c. For what time range is the acceleration negative?

In watter tydinterval is die versnelling negatief?

[2]

$$a = \frac{dv}{dt} = -30t$$

When t is positive, i.e. $t > 0 \text{ s}$, the acceleration will be negative.

Note that it is possible for time to be negative – this just means that this represents time before the chosen $t = 0 \text{ s}$.

2 Question 2 / Vraag 2

When the acceleration is constant, the average and instantaneous acceleration are equal:

$a = a_{avg} = \frac{v - v_0}{t - 0}$, where a is the acceleration, v is the velocity at time t and v_0 is the initial velocity

at $t = 0$ s. We can also write the average velocity as $v_{avg} = \frac{x - x_0}{t - 0}$, where x is the position at time t

and x_0 is the initial position at $t = 0$ s. Using these two equations show that when a particle is accelerating in the x direction, its displacement is given by $x - x_0 = v_0 t + \frac{1}{2} a t^2$. Show all steps and state all reasoning clearly.

Wanneer die versnelling konstant is, is die gemiddelde en oombliklike versnelling dieselfde:

$a = a_{gem} = \frac{v - v_0}{t - 0}$ waar a die versnelling is, v die snelheid by tyd t en v_0 die aanvanklike snelheid by

$t = 0$ s is. Ons kan ook die gemiddelde snelheid as $v_{gem} = \frac{x - x_0}{t - 0}$ skryf, waar x die posisie by tyd t

en x_0 die aanvanklike posisie by $t = 0$ s is. Deur gebruik te maak van hierdie twee vergelykings, bewys dat $x - x_0 = v_0 t + \frac{1}{2} a t^2$. Wys al die stappe en verduidelik jou aannames.

[7]

$$a = a_{avg} = \frac{v - v_0}{t - 0}$$

$$v = v_0 + at \quad (1)$$

$$v_{avg} = \frac{x - x_0}{t - 0}$$

$$x = x_0 + v_{avg} t \quad (2)$$

$$\text{But } v_{avg} = \frac{1}{2}(v + v_0). \quad (3)$$

Substituting this for v_{avg} in (2)

$$x = x_0 + \left[\frac{1}{2}(v + v_0) \right] t$$

From (1), we substitute for v into above and rearrange the terms:

$$\begin{aligned} x - x_0 &= \frac{1}{2} v t + \frac{1}{2} v_0 t \\ &= \frac{1}{2} (v_0 + at) t + \frac{1}{2} v_0 t \\ &= v_0 t + \frac{1}{2} a t^2 \end{aligned}$$

Since it was not specified – this derivation could also have been done using integration methods as shown in Ch 2.

(Ch 2 P 54)

3 Question 3 / Vraag 3

A stone is dropped into a river from a bridge 53.6 m above the water. Another stone is thrown vertically down 1.00 s after the first is dropped. The stones strike the water at the same time. Determine the initial speed of the second stone.

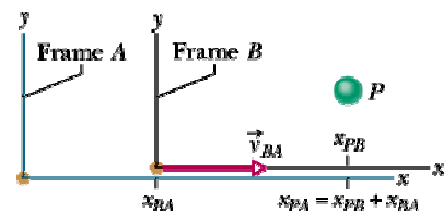
'n Klip word laat val in 'n rivier vanaf 'n brug wat 53.6 m bokant die water is. 'n Ander klip word vertikaal van die brug afgegooi, 1.00 sekondes nadat die eerste klip laat val is. Die twee klippe tref gelyk die water (op dieselfde tyd). Bereken die aanvanklike snelheid van die tweede klip.

[6]

Stone 1: $\Delta y = -53.6 \text{ m}$ $v_{0y1} = 0 \text{ m.s}^{-1}$ $a = -9.8 \text{ m.s}^{-2}$ $t = t_1$
$\Delta y = v_{0y1}t + \frac{1}{2}a_y t_1^2$ -----①
From ① $t_1 = \sqrt{\frac{2\Delta y}{a_y}}$ (= 3.31 s)
Stone 2: : $\Delta y = -53.6 \text{ m}$ $v_{0y2} = ??$ $a = -9.8 \text{ m.s}^{-2}$ $t_2 = t_1 - 1$
$\Delta y = v_{0y2}t + \frac{1}{2}a_y t_2^2$ -----②
For stone 2: $t_2 = t_1 - 1$ (= 2.31 s)
From ②: $v_{oy} = \frac{\Delta y - \frac{1}{2}at_2^2}{t_2}$
Substituting for t_2 : $v_{oy2} = \frac{\Delta y - \frac{1}{2}a \left(\sqrt{\frac{2\Delta y}{a_y}} - 1 \right)^2}{\sqrt{\frac{2\Delta y}{a_y}} - 1}$ or $v_{oy2} = \frac{(-53.6 \text{ m}) - \frac{1}{2}(-9.8 \text{ m.s}^{-2})(2.31 \text{ s})^2}{2.31 \text{ s}}$ $= -11.88 \text{ m.s}^{-1}$
$v_{oy} = -11.88 \text{ m/s}$

4 Question 4 / Vraag 4 (Ch 4 Sample prob)

Suppose that Alex (at the origin in frame A in the figure) is parked by the side of the road watching car P. Barbara (at the origin of frame B) is also watching car P, while traveling at a constant velocity relative to Alex, $v_{BA} = 52 \text{ km/h}$. Car P is moving in the negative direction of the x axis.



Veronderstel dat Arno (by die oorsprong van verwysingraamwerk A in die figuur) langs die pad staan en hou motorkar P dop. Bettie (by die oorsprong van verwysingsraamwerk B) hou ook kar P dop terwyl sy teen 'n konstante snelheid, $v_{BA} = 52 \text{ km/h}$, relatief tot Arno beweeg. Kar P beweeg in die negatiewe rigting van die x-as.

a. If Alex measures a constant velocity $v_{PA} = -78 \text{ km/h}$ for car P, what velocity will Barbara measure?

Indien Arno 'n konstante snelheid $v_{PA} = -78 \text{ km/h}$ meet vir kar P, wat is die snelheid gemeet deur Bettie?

[3]

v_{BP} is the velocity of P relative to Barbara/Bettie

$$v_{AP} = v_{AB} + v_{BP}$$

$$v_{BP} = v_{AP} - v_{AB}$$

$$= -78 \text{ km/h} - 52 \text{ km/h}$$

$$= -130 \text{ km/h}$$

b. If car P brakes and comes to a stop relative to Alex in time $t = 10 \text{ s}$ at a constant acceleration, what is its acceleration a_{PB} relative to Barbara?

Indien kar P rem en tot stiltand kom relatief tot Arno in 'n tyd $t = 10 \text{ s}$ teen 'n konstante versnelling, wat is die versnelling a_{PB} relatief tot Bettie?

[4]

If P is stationary relative to Alex – Barbara is still moving at $+52 \text{ km/h}$ relative to Alex and therefore to P .

Acceleration relative to Barbara:

$$\begin{aligned} a_{BP} &= \frac{\Delta v}{\Delta t} \\ &= \frac{52 \text{ km/h} - 130 \text{ km/h}}{(10 \text{ s} / 3600 \text{ s/h})} = -2,8 \times 10^4 \text{ km/h}^2 \\ &= \frac{14.4 \text{ m/s} - 36.1 \text{ m/s}}{(10 \text{ s})} = -2.17 \text{ m/s}^2 \end{aligned}$$

OR

Acceleration relative to Barbara is the same as the acceleration relative to Alex, which is:

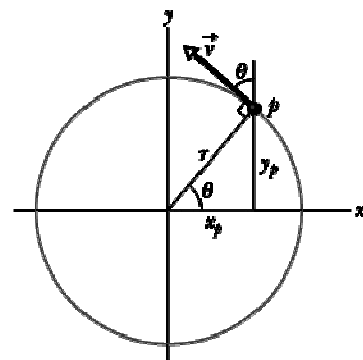
$$\begin{aligned} a_{BP} &= a_{AP} = \frac{\Delta v_{AP}}{\Delta t} \\ &= \frac{0 \text{ km/h} - 78 \text{ km/h}}{(10 \text{ s} / 3600 \text{ s/h})} = -2,8 \times 10^4 \text{ km/h}^2 \\ &= \frac{0 \text{ m/s} - 21.7 \text{ m/s}}{(10 \text{ s})} = -2.17 \text{ m/s}^2 \end{aligned}$$

5 Question 5 / Vraag 5

(Derivation) Consider a particle P , which is moving at a constant speed v , around a circle of radius r . At the instant shown, P has co-ordinates x_p and y_p . Stating all assumptions, and showing all your steps, show that the magnitude of the

acceleration of particle P is given by $a = \frac{v^2}{r}$.

Beskou 'n deeltjie P , wat met 'n konstante spoed v in 'n sirkelbaan met radius r beweeg. Op die tydstop aangedui in die figuur het P die koördinate x_p en y_p . Wys dat die grootte van die versnelling van deeltjie P is $a = \frac{v^2}{r}$. Dui al die stappe aan en



verduidelik al jou aannames.

[6]

coordinates: x_p, y_p

position: $\vec{r}$ is at angle θ to x-axis

$\vec{v} \perp$ position

$\vec{v}$ is at angle θ to perpendicular at P

$$\vec{v} = -v \sin \theta \hat{i} + v \cos \theta \hat{j}$$

$$= \left(-v \frac{y_p}{r} \right) \hat{i} + \left(v \frac{x_p}{r} \right) \hat{j}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \left(-\frac{v}{r} \left(\frac{dy_p}{dt} \right) \right) \hat{i} + \left(\frac{v}{r} \left(\frac{dx_p}{dt} \right) \right) \hat{j}$$

$$= \left(-\frac{v^2}{r} \cos \theta \right) \hat{i} + \left(\frac{v^2}{r} \sin \theta \right) \hat{j}$$

$$\frac{dy_p}{dt} = v_{yp} = v \sin \theta \quad \text{and} \quad \frac{dx_p}{dt} = v_{xp} = v \cos \theta$$

$$a = \sqrt{a_x^2 + a_y^2} = \frac{v^2}{r} \sqrt{\cos^2 \theta + \sin^2 \theta} = \frac{v^2}{r}$$

6 Question 6 / Vraag 6

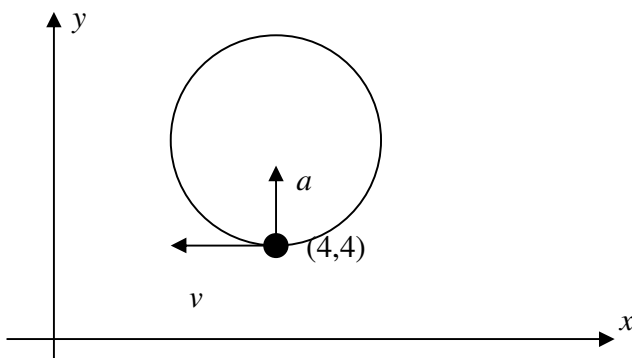
(Ch 4 P 64) A particle moves with uniform circular motion, in a horizontal xy plane. At one instant, it moves through point at coordinates (4.00m; 4.00m) with a velocity of $-5.00\hat{i} \text{ m/s}$ and an acceleration of $+12.5.00\hat{j} \text{ m/s}^2$.

'n Deeltjie beweeg met uniforme sirkelbeweging, in 'n horisontale xy vlak. Op 'n seker oomblik, beweeg die deeltjie deur die punt met y koördinate (4.00m; 4.00m) met 'n snelheid van $-5.00\hat{i} \text{ m/s}$ en versnelling van $+12.5.00\hat{j} \text{ m/s}^2$.

a. Draw the sketch showing clearly the orientation of the particle with respect to the centre of the circle, as well as the directions of the velocity and the acceleration vectors.

Teken 'n skets en dui duidelik die oriëntasie van die deeltjie ten opsigte van die middelpunt van die sirkel aan, sowel as die rigting van die snelheid en die versnelling vektore.

[3]



b. Determine the x and y coordinates of the centre of the circular path.

Bepaal die x en y koördinate van die middelpunt van die sirkelbaan.

[4]

At the point given, the particle has the same x coordinate as the center of the circle. The y coordinate of the centre is upward along the y -axis by a distance equal to the radius. The radius can be determined as follows:

$$a = \frac{v^2}{r}$$

$$r = \frac{v^2}{a}$$

$$= \frac{(5.00 \text{ m/s})^2}{12.5 \text{ m/s}^2}$$

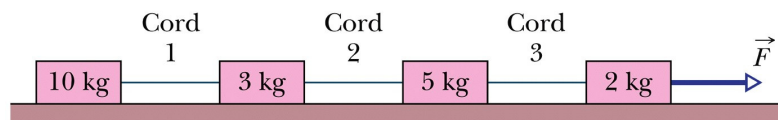
$$= 2.00 \text{ m}$$

Therefore the centre of the circular path is (4.00 m, 6.00 m)

7 Question 7 / Vraag 7 (Ch 5 Q7+e)

The figure shows four blocks being pulled to the right across a frictionless floor by force $\vec{F}$.

Die figuur dui vier blokke aan wat na regs getrek word oor 'n wrywinglose vloer met 'n krag $\vec{F}$.



a. What total mass is accelerated to the right by cord 3?

Wat is die totale massa wat deur tou 3 na regs versnel word?

[2]

18 kg

b. Rank the cords according to their tensions – greatest to smallest.

Rangskik die blokke volgens hul spankragte – grootste na kleinste.

[2]

3, 2, 1

c. If the magnitude of the force is $\vec{F} = 55 \text{ N}$, calculate the system's acceleration.

Indien die grootte van die krag $\vec{F} = 55 \text{ N}$ is, bereken die stelsel se versnelling.

[2]

Total mass of system: $M = 20 \text{ kg}$

Net external force on mass M :

$$\sum \vec{F} = Ma = 55 \text{ N}$$

$$a = \frac{F_{\text{net}}}{M} = \frac{55 \text{ N}}{20 \text{ kg}}$$

$$= 2.75 \text{ m.s}^{-2}$$

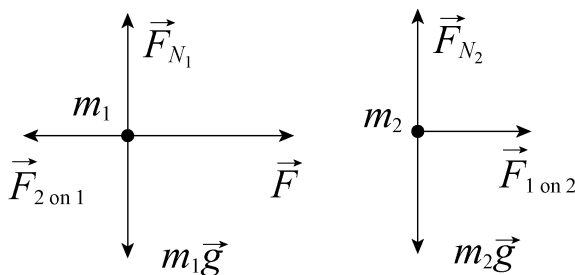
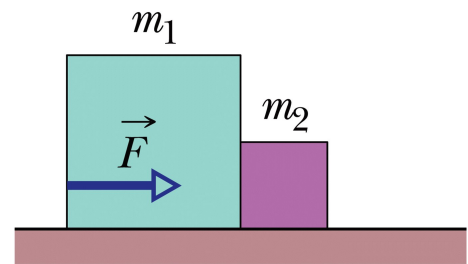
8 Question 8 / Vraag 8

Ch 5 P 55 – similar to tut problem 56 Two blocks are in contact on a frictionless table. A horizontal force is applied to the larger block as shown in the figure. If $m_1 = 2.3 \text{ kg}$, $m_2 = 1.2 \text{ kg}$ and $F = 3.2 \text{ N}$, answer the following questions.

Twee blokke is in kontak met mekaar op 'n wrywinglose tafel. 'n Horisontale krag word op die groter blok toegepas, soos aangedui in die figuur. Indien $m_1 = 2.3 \text{ kg}$, $m_2 = 1.2 \text{ kg}$ en $F = 3.2 \text{ N}$, beantwoord die volgende vrae.

a. Determine the magnitude of the force between the two blocks.

Bereken die grootte van die krag tussen die twee blokke.



[2 + 6]

$\vec{F}_{2\text{on}1} = -\vec{F}_{1\text{on}2}$ on block 1 (taking Newton's third law into account).

Newton's second law for block 1: $\sum F_1 = F - F_{2\text{on}1} = m_1 a$ ①

The second law for block 2: $\sum F_2 = F_{1\text{on}2} = m_2 a$ ②

Since the blocks move together they have the same acceleration.

From ② we obtain the expression

$$a = \frac{F_{1\text{on}2}}{m_2} \text{ (Cannot determine value yet),}$$

Substitute into ①

$$\sum F_1 = F - F_{2\text{on}1} = m_1 a$$

$$F - F_{2\text{on}1} = \frac{m_1 F_{1\text{on}2}}{m_2}.$$

Since $F_{2\text{on}1} = F_{1\text{on}2}$ (same magnitude for the third-law force pair), we obtain

$$F_{2\text{on}1} = F_{1\text{on}2} = \frac{m_2}{m_1 + m_2} F = \frac{1.2 \text{ kg}}{2.3 \text{ kg} + 1.2 \text{ kg}} (3.2 \text{ N}) = 1.1 \text{ N}.$$

Or you could have determined the net acceleration of the system $(m_1 + m_2)$ due to the net force and then determined the interaction force:

$$F_{net,x} = F = Ma$$

$$a = \frac{F}{M} = \frac{3.2}{3.5} = 0.914 \text{ m.s}^{-2}$$

For m_2 :

$$\sum F_x = m_2 a = F_{1 \text{ on } 2}$$

$$\begin{aligned} F_{1 \text{ on } 2} &= (1.2 \text{ kg})(0.914 \text{ m.s}^{-2}) \\ &= 1.097 \text{ N} \\ &\approx 1.1 \text{ N} \end{aligned}$$

b. If the same force is now applied to the smaller block, but in the opposite direction, without doing the calculation, will the force exerted by block 2 on block 1 be greater than, less than or equal to the value obtained in (a)? Give reasons for your answer.

Indien dieselfde krag nou toegepas word op die kleiner blok maar in die teenoorgestelde rigting, sonder om die berekening te doen, sal die krag wat blok 2 uitoefen op blok 1 groter as, kleiner as of gelyk aan die krag bereken in (a) wees? Gee redes vir jou antwoord.

[4]

The force will be greater.

We note that the acceleration of the blocks is the same in the two cases.

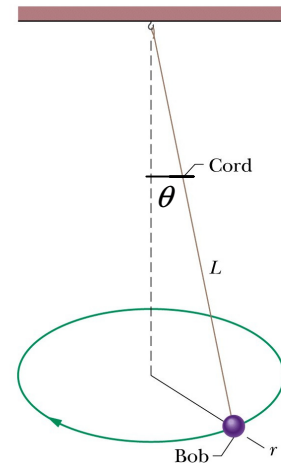
In part (a), the force $F_{1 \text{ on } 2}$ is the only horizontal force on the block of mass m_2 and in part (b) $F'_{2 \text{ on } 1}$ is the only horizontal force on the block with $m_1 > m_2$.

Since $F_{1 \text{ on } 2} = m_2 a$ in part (a) and $F'_{2 \text{ on } 1} = m_1 a$ in part (b), then for the accelerations to be the same, $F'_{2 \text{ on } 1} > F_{1 \text{ on } 2}$, that is, the force between the blocks must be larger in part (b).

9 Question 9 / Vraag 9

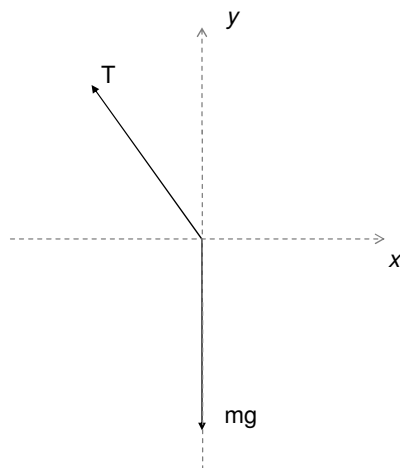
(Ch 6 P 70) The figure shows a conical pendulum, in which the bob (the small object at the lower end of the cord) moves in a horizontal circle at constant speed. (The cord sweeps out a cone as the bob rotates.) The bob has a mass of 0.050 kg, the string has a length $L = 0.90$ m and has negligible mass, and the bob follows a circular path of circumference 0.94 m.

'n Koniese slinger waar die sferiese massa (aan die punt van die tou) in 'n horisontale sirkel beweeg teen 'n konstante spoed, word in die figuur aangedui. (Die tou beweeg so, dat dit 'n keël vorm wanneer die massa roteer) Die massa is 0.050 kg, die tou het 'n lengte van $L = 0.90$ m met 'n weglaatbare massa, en die massa beweeg in 'n sirkelpad met 'n omtrek van 0.94 m.



- a. Draw the free-body diagram of the bob and determine the tension in the cord.

Teken die vryligaams diagram van die sferiese massa en bepaal die spankrag in die tou.



[2+6]

$m = 0.050$ kg	$L = 0.90$ m	circumference = 0.94 m
Acceleration is horizontal and is directed towards the centre of the circular path of the mass (bob).		

Determine the radius of the circle:

$$\text{circumference} = 2\pi R$$

$$R = 0.94/2\pi = 0.15 \text{ m.}$$

Determine the angle θ .

The angle that the cord makes with the **horizontal** is now easily found:

$$\theta = \sin^{-1}(R/L) = \sin^{-1}(0.15 \text{ m}/0.90 \text{ m}) = 9.6^\circ \approx 10^\circ$$

The vertical component of the force of tension in the string is $T\cos\theta$ and must equal the downward pull of gravity (mg). Thus the tension T can be determined:

$$\sum F_y = ma = 0 = T \cos \theta - mg$$

$$T = \frac{mg}{\cos \theta} = 0.50 \text{ N}$$

b. Determine the net acceleration of the bob.

Bepaal die netto versnelling van die massa.

[3]

The forces acting on the bob in the horizontal direction:

$$\sum F_x = ma = T_x = T \sin \theta$$

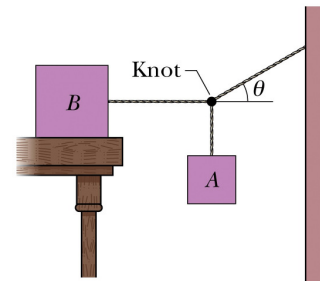
$$a = \frac{T \sin \theta}{m} = \frac{(0.50 \text{ N}) \sin \theta}{0.050 \text{ kg}}$$

$$a = 1.7 \text{ m.s}^{-2}$$

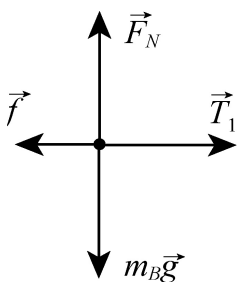
10 Question 10 / Vraag 10 (Ch 6 p 25- homework)

Block B in the figure weighs 711 N. The coefficient of static friction between block and table is 0.25; the angle θ is 30° ; assume that the cord between B and the knot is horizontal. Find the maximum weight of block A for which the system will be stationary.

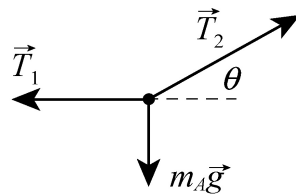
Die gewig van blok B in die figuur is 711 N. Die statiese wrywingskoëffisiënt tussen die blok en die tafel is 0.25. Die hoek θ wat aangedui is, is 30° . Veronderstel dat die tou tussen B en die knoop horisontaal is. Bereken die maksimum gewig van blok A, sodat die stelsel nie beweeg nie.



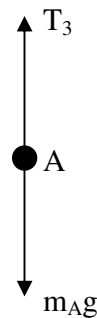
[2+9]



Block B



The knot



Block A

$\vec{T}_1$ is the tension force of the rope pulling on block B or pulling on the knot (as the case may be),
 $\vec{T}_2$ is the tension force exerted by the second rope (at angle $\theta = 30^\circ$) on the knot,
 $\vec{T}_3$ is the tension force exerted by block A on the knot (and is equal to $m_A g$ (when applying Newton's second law to block A).

$\vec{f}$ is the force of static friction exerted by the horizontal surface on block B,

$\vec{F}_N$ is normal force exerted by the surface on block B,

W_A is the weight of block A (W_A is the magnitude of $m_A \vec{g}$), and W_B is the weight of block B ($W_B =$

711 N is the magnitude of $m_B \vec{g}$).

Applying NII law to block B and knot:

Block B

$$\sum F_{Bx} = T_1 - f_s = m_B a_x = 0$$

$$\sum F_{By} = N_B - m_B g = m_B a_y = 0$$

$$T_1 = f_s = \mu_s N_B$$

$$N_B = m_B g$$

Then :

$$T_1 = \mu_s N_B = \mu_s m_B g$$

the knot:

$$\sum F_{knot,x} = -T_1 + T_2 \cos \theta = m_{knot} a_x = 0$$

$$\sum F_{knot,y} = T_2 \sin \theta - m_A g = m_{knot} a_y = 0$$

$$-T_1 + T_2 \cos \theta = 0$$

$$T_2 = \frac{T_1}{\cos \theta}$$

Substituting :

$$W_A = m_A g = T_2 \sin \theta = \frac{T_1 \sin \theta}{\cos \theta}$$

$$= \frac{\mu_s m_B g \sin \theta}{\cos \theta}$$

$$= 102.6 = 103 \text{ N}$$

$$\approx 1.0 \times 10^2 \text{ N}$$

THE END / DIE EINDE